M2 Mid-Submission Report

Accessible AI for All: Shopping Track (omnikart Agent)

Team: 289

1. Current Approach & Changes Since M1

Our core architecture remains intact: we are building "omnikart", a conversational, voice-first shopping assistant designed to simplify e-commerce for users with visual, motor, and age-related impairments. Based on integration testing and API constraints, we have implemented four functional deviations from our M1 plan.

ASR Confidence Replaced

Because AnuvaadHub does not return ASR confidence scores (planned as a 0.62 threshold), we now confirm lone numbers explicitly and require an exact phrase match ("Yes, place order") to authorize checkout.

Interruption Mechanism

Talking over the "omnikart" agent proved unreliable via Voice Activity Detection (VAD). Users now interrupt the AI using a deliberate physical key press or screen tap.

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Review Summarization

The four "gstack" plan reviews were converted into static evaluation checklists rather than running as dynamic, full-question sessions to save API bandwidth and time.

Optional Features

The offline voice pack and Groq-driven review summaries have been decoupled. They are now strictly optional fallbacks, ensuring the core app remains functional without them.

2. Experiments Run & Current Results

We have successfully established the foundational pipeline. All model APIs are actively receiving requests

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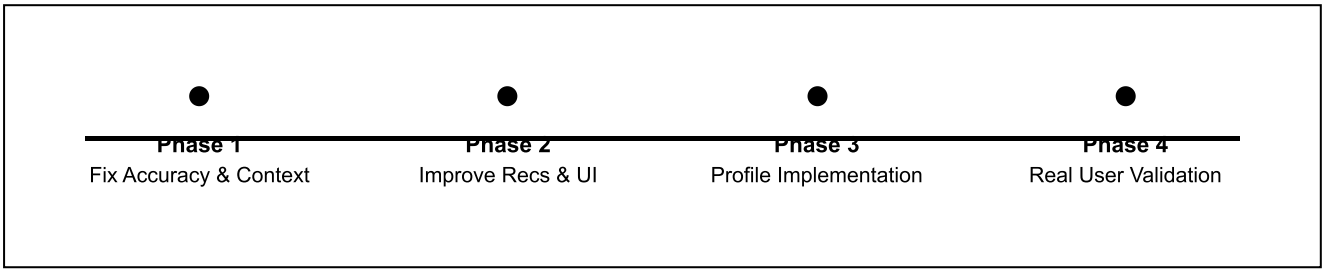
3. Progress on Testing & Issue Mitigation

We have shifted from unit testing individual nodes to refining the user experience layer. We are currently actively addressing the bottlenecks identified during our initial integration experiments to ensure a seamless checkout flow.

Identified Issue	Current Progress / Status	Mitigation Strategy
Limited Mock Web	Building a more realistic mock e-commerce environment to simulate real DOM structures.	Expanding catalog depth and injecting simulated noise.
Missing User Setup	Working on a configuration flow to determine the user's specific disability category.	Implementing dynamic parameter selection at launch.
LLM Context Loss	Working on better recommendations and memory retention for the LLM agent.	Condensing conversation history into strict JSON payloads.
API Latency	Pipeline is functional but slow during cross-language turns.	Optimizing prompt sizes and parallelizing independent API calls.

4. Plan for the Remaining Time

Our focus for the final sprint shifts heavily toward accuracy, interface polish, and human validation. We must transition the application from a functional technical prototype to a highly user-friendly assistant.



The final crucial step before the final submission deadline is **interacting with a real disabled person**. Validating our assumptions regarding screen reader redundancy, turn-taking cadence, and simplified checkout against actual user behavior will finalize our design parameters and inform our final presentation.